

# HSBC: Recharge Failure Recovery System

Developed an architecture using the circuit breaker pattern to ensure system resilience during external service outages. Key technologies used:

- AWS
- Kubernetes
- Microservices
- PostgreSQL
- Cassandra
- Kafka (asynchronous communication)

**Problem Statement:** external services were down and users were unable to do successful recharges

**Solution:** To solve this issue, we designed the system to show users the bill details and process the recharge request once the services became available.

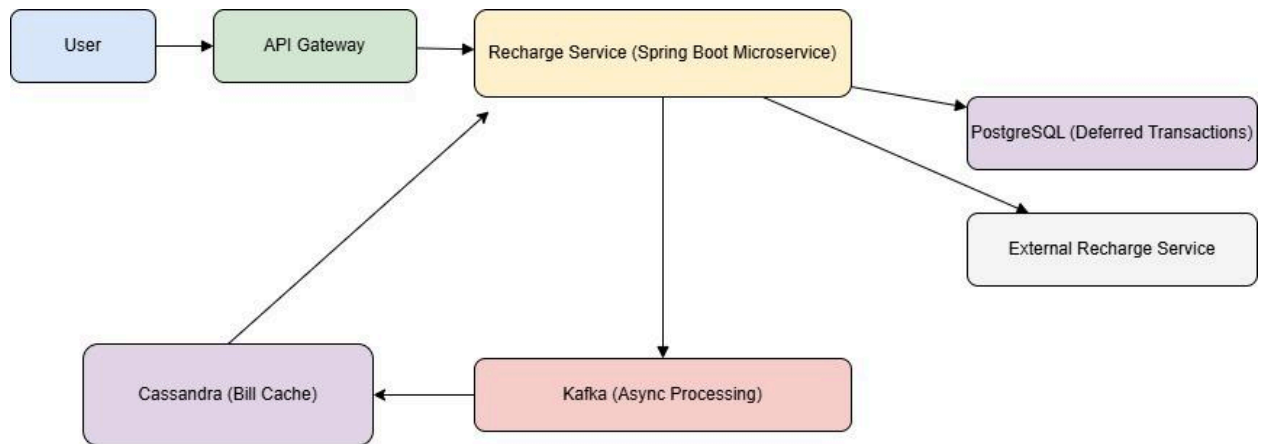
## Three-Stage Recovery Process:

1. **Data Caching:** Bill details from external services are cached in Cassandra during normal operations.
2. **Failure Handling:** during failures, we fetched these bill details from Cassandra and stored the recharge requests in PostgreSQL in a deferred state.
3. **Recovery & Processing:** we validated whether the external service was up or not by sending a few test requests. Once the service became available, we started processing the deferred requests successfully.

## Circuit Breaker States

| State            | Description                                                                                                                                           |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Closed</b>    | Normal operations with direct external service calls.                                                                                                 |
| <b>Open</b>      | External service failures occurred, so we avoided repeated external calls and used Cassandra data while storing recharge requests in a deferred state |
| <b>Half-Open</b> | we validated service recovery by sending a few requests to the external services                                                                      |

# Architecture



## Circuit-Breaker Diagram

